MAINTAINER REFERENCE

# P U L S E

## Technical Reference

Architecture, SharePoint integration, packaging, deployment, verification, and maintainer handoff for the PULSE operating workspace.

| AUDIENCE | SCOPE | USE |
| --- | --- | --- |
| Developers and maintainers | Current source package | Internal reference |

00 / DOCUMENT AUTHORITY

# Built for the next maintainer.

This reference organizes the verified architecture and operating constraints of PULSE so a new maintainer can trace a change from the user interface to SharePoint, build an approved artifact, validate it safely, and recover when acceptance fails.

SOURCE HIERARCHY

### Current implementation first.

Use the current repository, build scripts, schema, and hosted behavior as the authority. Historical Firepit and Forge notes remain useful where they describe active paths, but do not assume the production host without verification.

KNOWN UNCERTAINTY

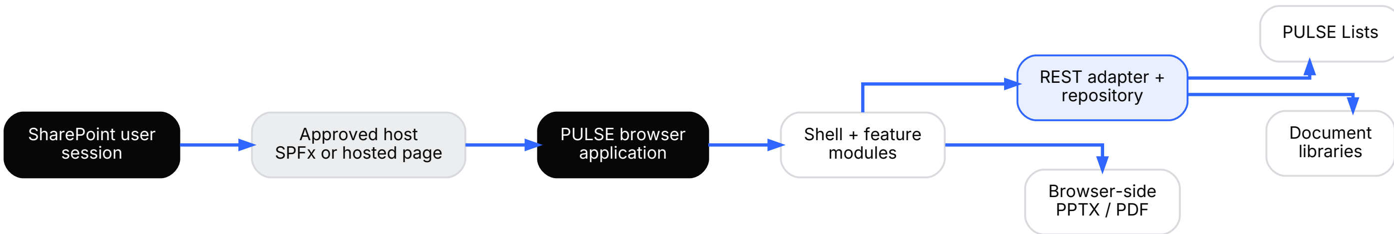
### Production delivery surface

The repository contains both a self-contained browser package and an SPFx host that embeds the browser implementation. Confirm which path is approved and deployed before changing release procedures.

01 / SYSTEM ARCHITECTURE

# Browser-first by design.

PULSE runs in the user's browser, relies on the existing SharePoint session, and uses SharePoint Lists and document libraries as its persistence layer. There is no separate PULSE application server, database server, or sign-in system.




**Compute**
Feature logic executes in the browser.

**Identity**
Current SharePoint session and role record.

**Storage**
PULSE Lists and project document libraries.

**Output**
Browser-side PowerPoint and PDF generation.

CORE INTERPRETATION

## Same session. Same boundary. No parallel platform.

The architecture deliberately minimizes infrastructure outside what the SharePoint site and the user's browser already provide.

02 / HOSTING AND SECURITY BOUNDARIES

# Two repository paths. One decision to verify.

The current repository contains a self-contained browser package and an SPFx host solution. The SPFx web part embeds the browser implementation through `LegacyWorkspaceFrame.tsx`; the typed SPFx page modules are not the active delivered feature surface.

BROWSER PACKAGE

### Self-contained HTML

Built from `apps/PULSE`. Local CSS, JavaScript, images, and approved browser libraries are packaged into one artifact for SharePoint or a compatible host.

SPFX HOST

### Embedded legacy workspace

The host solution in `apps/spfx-pulse-workspace` renders the browser application. Treat it as a delivery shell, not a parallel full implementation.

## Security and operating constraints

| Boundary | Maintainer implication |
| --- | --- |
| **Existing SharePoint session** | Do not add a separate login or duplicate identity model. |
| **Same-site REST** | Write operations use the current session and request digest. Validate in a real SharePoint context. |
| **No secret material in source** | Runtime configuration may contain non-secret settings only. Never place tokens, credentials, or private addresses in the package. |
| **Self-contained delivery** | The approved browser artifact must not depend on external runtime styles or scripts. |
| **Production host [VERIFY]** | Confirm the approved deployed path before updating release or rollback instructions. |

**Maintainer rule:** do not resolve the hosting uncertainty by inference. Verify the real deployed surface and release authority with the system owner.

03 / REPOSITORY MAP

# Know where the behavior lives.

The active feature surface remains the browser implementation under `apps/PULSE`. The SPFx solution provides the current repository's SharePoint Framework integration path.

| Path | Responsibility |
|---|---|
| `apps/PULSE/index.html` | Browser entry point and dependency order. |
| `apps/PULSE/assets/js/app.js` | Boot, shell, hash routing, shared UI helpers, and diagnostics. |
| `apps/PULSE/assets/js/data.js` | Canonical browser store, normalization, and local fallback. |
| `apps/PULSE/assets/js/pages/` | Module-specific pages and workflows. |
| `apps/PULSE/assets/js/sharepoint-adapter.js` | Same-site REST transport, site detection, identity, and role operations. |
| `apps/PULSE/assets/js/sharepoint-repo.js` | Object-to-list mapping, persistence, and removal. |
| `apps/PULSE/assets/js/sharepoint-schema.js` | List definitions, validation, provisioning, and field repair. |
| `apps/PULSE/assets/js/*pptx-export.js` | Client-side project and overview PowerPoint output. |
| `apps/PULSE/scripts/build-sharepoint-package.js` | Preferred self-contained browser-package builder. |
| `apps/spfx-pulse-workspace/` | SPFx host solution and browser-app integration. |

EDIT PRIORITY

1. Keep SharePoint transport working.
2. Keep identity and role resolution working.
3. Keep setup and schema validation working.
4. Keep diagnostics usable.
5. Then change feature UI.

AVOID DUPLICATION

Route SharePoint operations through the adapter and repository. Do not scatter raw REST calls or create a second mapping layer inside a page module.

04 / CONFIGURATION

# Configuration is not a secret store.

`apps/PULSE/assets/js/app-config.js` holds non-secret runtime settings such as list prefix, site-detection fallback, host-page naming, group fallbacks, diagnostics, and the government Graph base URL.

ALLOWED

### Environment behavior

- List naming and prefixes
- Non-secret site-detection fallback
- Host-page route names
- Government Graph base URL
- Diagnostics and feature flags

NOT ALLOWED

### Credentials or private data

- API keys or access tokens
- Cookies or request digests
- Personal accounts
- Private production URLs in distributable docs
- Classified or controlled operational records

## Site and identity resolution

01

### Page context

Prefer SharePoint's injected page context when available.

02

### Configured fallback

Use only an approved non-secret manual site value.

03

### Local fallback

When no site is detected, load non-persistent local state for UI work only.

**AI review [VERIFY]:** the source package does not establish an approved endpoint, key-management path, or authorization to operate. Do not enable or document it as an operational capability until those controls are confirmed.

05 / BUILD AND PACKAGING

# Build the artifact the host expects.

The preferred browser-package builder reads the unbundled entry point, inlines approved local dependencies, rewrites local asset references, removes development cache strings, and rejects external runtime styles or scripts.

PREFERRED BROWSER PACKAGE

```
cd apps/PULSE
node scripts/build-sharepoint-package.js "releases/PULSE-YYYY.MM.DD.html"
```

### What the builder should preserve

- Dependency order from `index.html`
- Local styles and scripts
- Embedded approved images
- Browser-global library registration
- Build provenance metadata

### What the builder should reject

- External runtime script URLs
- External runtime stylesheets
- Unresolved local assets
- Development-only cache query strings
- Output without a deliberate destination

SPFX HOST VALIDATION

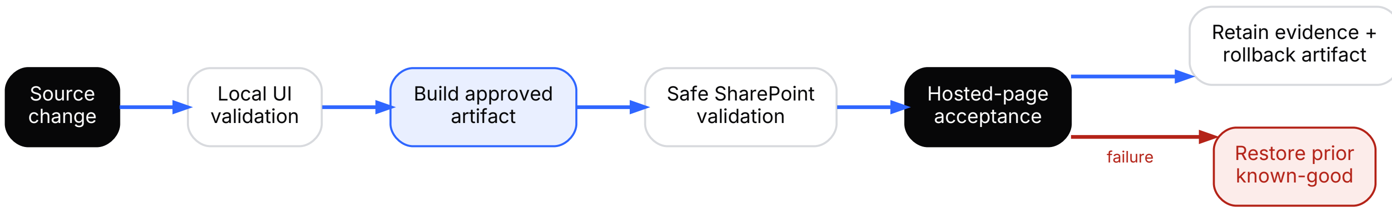
```
cd apps/spfx-pulse-workspace
./use-node22.sh npm install
./use-node22.sh gulp serve
./use-node22.sh npm run typecheck
./use-node22.sh npm run lint
./use-node22.sh npm run build
```

**Do not substitute build paths casually.** The browser artifact and SPFx package serve different roles. Confirm the approved release path before shipping either one.

06 / DEPLOYMENT AND ROLLBACK

# Acceptance happens in SharePoint.

A local browser proves layout and client logic. It does not prove identity, role association, list access, persistence, document-library behavior, or hosted-page restrictions.




**BEFORE UPLOAD**

- Retain the prior known-good artifact.
- Record the source revision and build time.
- Complete local UI checks.
- Confirm the target and release authority.

**AFTER UPLOAD**

- Open the real hosted page.
- Validate identity and role resolution.
- Check navigation and diagnostics.
- Perform a controlled read and write for changed record types.

### Rollback decision

If acceptance fails, restore the prior known-good artifact first. Preserve the failed artifact, browser console evidence, REST response, affected route, user role, and reproduction steps. Investigate from source after service is restored.

**DEFINITION OF DONE**

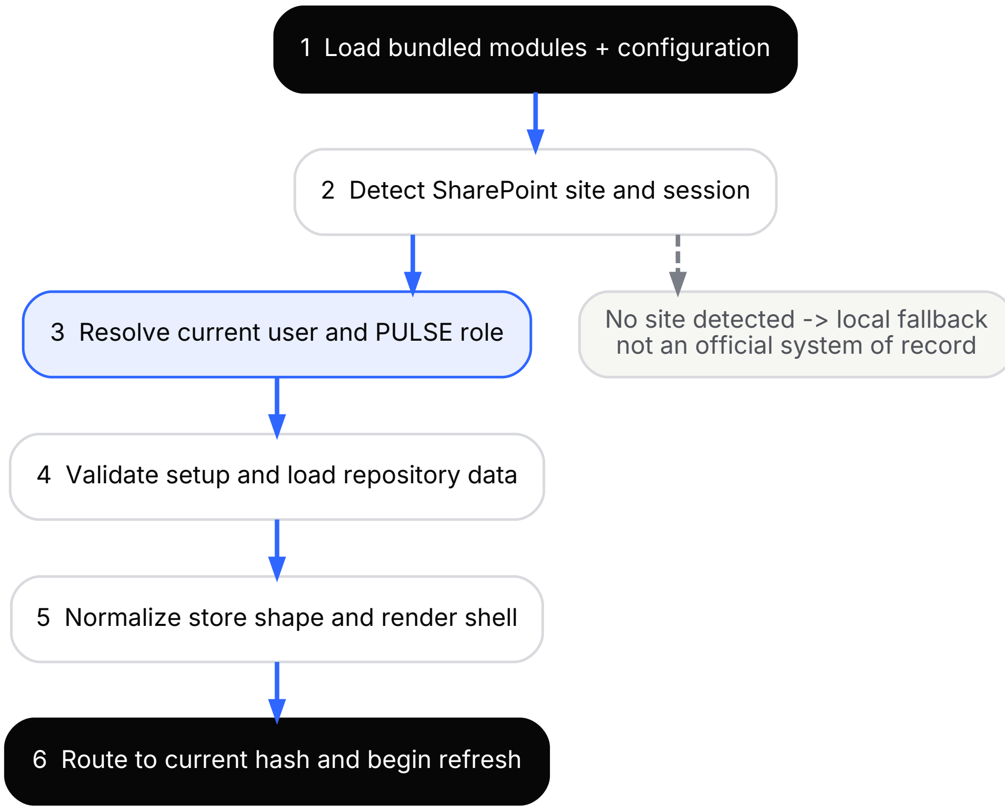
## Built. Hosted. Read. Written. Recoverable.

A release is not complete until the real hosted environment accepts the artifact and a rollback path remains available.

07 / BOOT SEQUENCE AND ROUTING

# Fail visibly, not silently.

The browser entry point loads vendor libraries, configuration, SharePoint integration, store normalization, reporting exports, support subsystems, the shell, and page modules before resolving the current route.




### ROUTING

Navigation is hash-based. The shell parses the route, selects the matching page renderer, and gates restricted routes before rendering into the page-content mount point.

### FAILURE BEHAVIOR

A top-level startup failure should show a plain diagnostic state rather than a blank page. Page-level rendering errors should remain contained to the affected route.

**Local fallback:** useful for UI development, but it is empty or browser-local and is not the official system of record.

08 / SHAREPOINT INTEGRATION

# One data boundary, handled deliberately.

All core reads and writes use the SharePoint REST API through the adapter and repository. Authentication comes from the current page session; write requests obtain a request digest.

READ PATH

### REST → mapper → normalized store

List items are hydrated into browser objects, normalized, and cached in the in-memory store used by page modules.

WRITE PATH

### UI → validation → mapper → REST

A visible save must not be treated as complete until the repository write resolves and the returned SharePoint identity is applied to the browser object.

## Maintainer invariants

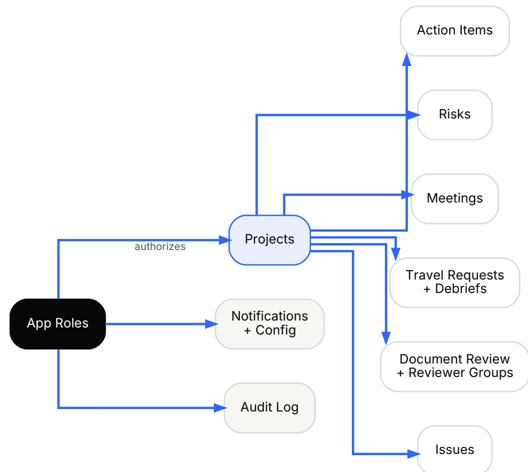
| Invariant | Reason |
|---|---|
| **Keep raw REST in the adapter** | Centralizes digest handling, diagnostics, people resolution, and transport behavior. |
| **Keep schema and mappers synchronized** | A UI field without a real SharePoint column can appear to save while the value is silently omitted. |
| **Use raw bytes for binary upload** | Text encoding corrupts Office documents and images. |
| **Test the live site** | Local mode cannot prove permissions, list readiness, or persistence. |
| **Do not infer app roles from site admin** | SharePoint administration is not the PULSE authorization model. |

**Schema changes are cross-cutting:** update the schema definition, write mapper, read mapper, normalization, UI validation, setup/provisioning, and live read/write test as one change.

09 / CURRENT SHAREPOINT SCHEMA

# Lists form the application record.

The current source package identifies the following list titles. Confirm exact production titles and document-library names on the approved SharePoint site before performing setup, migration, or recovery.




**PULSE App Roles**
User association and authorization.

**PULSE Projects**
Project portfolio and project-level metadata.

**PULSE Risks**
Project risks and mitigation records.

**PULSE Action Items**
Tasks and action records.

**PULSE Meetings**
Global and project meeting sessions.

**PULSE Team Events**
Team-level event records.

**PULSE Travel Requests**
Travel and leave workflow records.

**PULSE Travel Debriefs**
After-action records linked to travel.

**PULSE Document Review**
Formal review workflow and history.

**PULSE Doc Reviewer Groups**
Reusable review routing groups.

**PULSE Issues**
Support ticket and blocker records.

**PULSE Notifications**
Outbound notification queue or records.

**PULSE Notification Config**
User and system notification settings.

**PULSE Audit Log**
Supported activity and change history.

**PULSE App Settings**
Single-row or shared application settings.

**PULSE Location Config**
Approved location and travel configuration.

> **Document storage is separate.** Project files use SharePoint document libraries. Formal Document Review lifecycle data remains in its dedicated list and must not be collapsed into file storage.

10 / DATA MODEL AND REPOSITORY PATTERN

# Map once. Normalize every load.

The repository converts SharePoint list items into a normalized browser store used by every page. Composite areas may be serialized into JSON fields when the application treats them as one record boundary.

## 01

### Load

Read lists, hydrate records, then normalize missing or older fields into the current store shape.

## 02

### Use

Page modules read from one in-memory store rather than calling SharePoint independently.

## 03

### Persist

Repository mappers serialize validated objects back to existing list columns.

### Cross-cutting field checklist

- Schema definition exists with the correct internal and display names.
- Read mapper hydrates the field back into the browser object.
- UI validation matches the real column type and allowed choices.
- Write mapper emits the field in the expected SharePoint format.
- Normalization provides a safe value for older or missing records.
- SharePoint setup has been run and live read/write verified.

CONCURRENCY

When a create returns the SharePoint item ID, later updates to the same object must wait for that result. Do not fire a second save against an object whose identity has not been stamped.

REFRESH PROTECTION

Background refresh must not overwrite an active edit or recently written record. Keep modal, pending-save, and recent-write guards intact.

11 / ROLES AND PERMISSIONS

# Authorization comes from PULSE App Roles.

After current-user resolution, PULSE associates the user with an active role record. Administrative rights, ordinary membership, and project access should be evaluated through the application role service.

PRIMARY AUTHORITY

### Active role record

Use the approved role association and the application's current matching logic. User administration remains restricted to administrators.

RECOVERY EXCEPTION

### SharePoint site admin

A site administrator may require setup or recovery access when role data is missing, but site-admin status is not a substitute for a PULSE role.

## Identity and role validation

| Check | Expected evidence |
|---|---|
| **Current user** | SharePoint REST identity agrees with the visible user context. |
| **Role association** | An active PULSE App Roles record matches the resolved user. |
| **Restricted navigation** | Admin and other gated routes remain hidden or blocked for ordinary users. |
| **Project access** | Project-scoped controls follow the current user's assignment and role. |
| **Recovery path** | Approved administrators can reach setup without granting ordinary feature rights. |

**Do not gate feature permissions directly on `IsSiteAdmin`.** That property belongs to SharePoint administration, not the PULSE business-role model.

12 / NOTIFICATIONS AND AUDIT

# Actions should route and leave evidence.

Notification preferences and system configuration control which PULSE areas can contact a user and through which channel. Supported create, update, delete, navigation, and workflow actions may be recorded in the audit log when persistence succeeds.

NOTIFICATION PAYLOAD

### One event, contextual delivery

- Recipient and preferred channels
- Application area and record context
- Direct route or deep link
- Human-readable subject and message
- Delivery state when queued

AUDIT PAYLOAD

### Who, what, where, when

- Actor and resolved role
- Action and application area
- Route and object type
- Relevant record context
- Timestamp and persistence result

## Maintainer checks

- Preferences are resolved for the intended user.
- Route parameters open the correct context.
- Audit entries do not contain secret material.
- The source record exists before a deep link is sent.
- Failure does not erase the source workflow action.
- Delivery and audit behavior are verified on SharePoint.

**Trust boundary:** a notification is a pointer, not the authoritative record. Users should confirm the source record inside PULSE before acting.

13 / REPORTING AND EXPORTS

# Generate in the browser. Verify before use.

Project and group PowerPoint output is generated client-side with the bundled PptxGenJS runtime. The export path reads current browser-store data, builds slides, and returns a file without a separate reporting server.

### 01 Assemble

Select projects or group context and normalize the reporting data.

### 02 Generate

Create slides and assets with the bundled browser library.

### 03 Deliver

Download locally or, where supported, upload the package to project files.

## Verification before distribution

| Area | Check |
|---|---|
| **Content** | Project identifiers, owners, dates, milestones, risk language, and current state are accurate. |
| **Selection** | The output contains only the intended project or group records. |
| **Images** | Photos and logos are present, proportioned, and safe to distribute. |
| **File** | The PowerPoint opens, remains editable where intended, and has no clipped elements. |
| **Destination** | If uploaded, the approved project library path and permissions are correct. |
| **Traceability** | The generated file can be connected to the source records and release context. |

**Export success is not content approval.** A generated file must be reviewed before it is used for an official brief or external distribution.

14 / TESTING AND VERIFICATION

# Local first. SharePoint final.

Use local development to shorten the UI feedback loop, then test the full change in a safe SharePoint environment. The second step is mandatory for any code that touches identity, roles, lists, files, notifications, or deployment.

LOCAL CHECKS

- Page loads without console errors
- Route renders the intended module
- Forms and modals remain usable
- Responsive layout remains stable
- Browser export generation completes

SHAREPOINT CHECKS

- Current user and role resolve correctly
- Required lists and columns exist
- Controlled create, update, and delete persist
- Document upload and open paths work
- Notifications and audit entries appear

## Recommended regression sequence

- Boot and diagnostics
- Role-restricted routes
- Action item and nested-data save
- Workflow notification
- Project PowerPoint
- Dashboard and navigation
- Project read and write
- Document-library operation
- Audit record
- Group PowerPoint

**Evidence:** retain the build artifact, source revision, test role, target site, changed record type, expected result, actual result, and any screenshots or REST responses needed to reproduce a failure.

15 / TROUBLESHOOTING

# Inspect the boundary before rewriting the feature.

Most high-impact failures occur at packaging, identity, schema, mapping, or hosted-environment boundaries. Diagnose those areas before changing page logic.

| Symptom | First inspection | Safe response |
| --- | --- | --- |
| **Blank or partial page** | Hosted page, console, generated package, asset paths, provenance | Rebuild with the preferred builder. Compare to the prior known-good artifact. |
| **Incorrect role** | Current user, active role record, diagnostics, target site | Correct the approved role record. Do not infer authorization from site membership. |
| **Save appears successful but data is missing** | Live column, mapper pair, filtered payload, REST response | Repair schema and mappings together, run setup, then repeat live read/write validation. |
| **Choice value will not persist** | UI options versus the SharePoint Choice column | Align schema and UI choices, provision the change, and retest. |
| **Binary file is corrupt** | Upload path and request body type | Use raw bytes. Do not route binary data through text encoding. |
| **Stale data overwrites an edit** | Background refresh, open modal, pending save, recent write | Restore refresh guards and isolate the timing before changing the store. |
| **Notification does not arrive** | User preferences, notification config, source event, delivery record | Confirm the source action first, then inspect routing and delivery status. |
| **Export fails or is malformed** | Browser console, PptxGenJS load, image data, selected record set | Reproduce with a minimal safe record, fix the export boundary, and visually inspect the file. |
| **Release-only defect** | Hosted page, target permissions, prior artifact | Roll back first, preserve evidence, then isolate the source change. |

DIAGNOSTIC ORDER

## Host → identity → role → schema → mapper → feature.

Following the boundary inward prevents a visible symptom from being mistaken for an architectural failure.

16 / CHANGE WORKFLOW AND HANDOFF

# Leave the system easier to maintain.

Every change should leave a clear path from request to source, build, hosted verification, release evidence, rollback, and updated documentation.

CHANGE WORKFLOW

1. Identify the active feature file and data boundary.
2. Update schema, mapper, normalization, and UI together when data changes.
3. Validate locally.
4. Build the approved artifact.
5. Test on a safe SharePoint site.
6. Deploy with a retained rollback copy.
7. Record evidence and update references.

MAINTAINER HANDOFF

- Approved source location and branch
- Production host and release authority [VERIFY]
- Approved SharePoint site and libraries [VERIFY]
- Known-good rollback artifact
- Current list schema and role records
- Open defects and technical debt
- Last successful test evidence

## Known limitations requiring confirmation

- The current production hosting arrangement is not available in the source package. Confirm whether the approved surface is the browser package, the SPFx host, or another host.
- Confirm the approved live SharePoint site, document-library names, retention controls, and deployment authority.
- Confirm whether AI grammar review has an approved endpoint, key-management path, and authorization before enabling it.
- The typed SPFx structure is not a complete standalone feature implementation because the host currently renders the embedded browser application.

FINAL MAINTAINER RULE

## Trace the change end to end.

UI → store → repository → schema → SharePoint → hosted verification → release evidence.